Property-Testing Strategic AI Routing Markets: An Audited Multi-Agent Environment

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Abstract

Inference routers allocate token demand among strategic providers with different capacity, cost, quality, and learning rules. We introduce a request-level multi-agent environment with exact settlement, capacity, fallback, common random numbers, provider-private feedback, pluggable routers, and adversarial deviation oracles. A property ladder separates software invariants, intervention fidelity, held-out market moments, strategic robustness, and live transport. On 3.18 million public endpoint observations, three GLM-5.2 active undercutters receive 18.62 percentage points of inverse-price shadow share from passive quotes, yet GLM effective rank is linear-compatible and 90% of non-GLM markets have no active undercutter. Thus the proposed asymptotic regime split fails its empirical property test. A fixed-n information-congestion benchmark recovers the theoretical optimizer to exponent error below 0.006, while its paired bandit intervention moves action correlation without moving allocation. A second game recomputes bounded best responses for ten provider technologies under 108 router rules: welfare-, revenue-, quality-, user-, and provider-oriented optima diverge, and pair deviations remain profitable despite zero unilateral grid regret. An owned price-sort intervention changes cheapest-provider selection by 80 points but remains preregistration-gated. The result is an executable refusal standard for mechanism claims, not a claim that simulated agents reproduce live providers or welfare.

1 Introduction

AI inference is becoming a strategic allocation problem. A harness sends a request naming an open-weight model; a router chooses among providers that run that model; and providers choose price, admission, and service quality. The objects are partly substitutable but not identical. Sampling, quantization, tool-call support, latency, privacy, and hidden capacity alter the delivered product even under a common model label.

This is a natural multi-agent learning environment. Providers repeatedly act, receive private profit feedback, and adapt against a router that transforms their actions into demand. It is also a dangerous environment to study naively. A public price menu is not a firm execution commitment. A documented inverse-price score is not necessarily realized selection after eligibility and fallback. A simulator can generate clean welfare and collusion statistics even when the corresponding live quantities are unobserved.

We propose a different standard: a routing simulator should be *property-tested*. Its internal invariants must hold exactly; its observations must match the information available to real agents; its calibration targets must survive holdout; causal simulator interventions must preserve declared nuisance structure; and a separate transport gate must pass before interpreting a simulated mechanism in the live market. Failed gates are results rather than inconveniences.

We implement this standard in `orcap`, a deterministic request-level kernel with a lightweight parallel multi-agent API. A provider action contains a quote, admitted-capacity fraction, and availability state. A router plugin returns an attempted-provider order. Settlement applies capacity constraints, technical failure, fallback latency, payment, variable cost, capital cost, user utility, and welfare request by request. Stable hash-derived substreams give common random numbers across mechanism arms. Provider observations expose public quotes and only the provider's own realized settlement; rival cost, capacity, and reliability remain private.

The environment includes random, inverse-price, quality-adjusted, synthetic price-cut-memory, menu-projected, and hardened adaptive routers. Provider policies include static regimes, calibrated species, tabular Q-learning, UCB, epsilon-greedy, and an optional LLM adapter. Global finite-grid best responses, two-provider coalitions, identity splitting, fading quotes, and sequential response oracles audit strategic robustness. The memory router is an adversarial benchmark with a declared transition law; it is not calibrated from the public price panel.

Four tasks are intentionally separated. E0 measures a contemporaneous price–score manipulation surface. E1 asks whether particular learners react to correlated signal order inside a synthetic game. E2 audits a hardened score against deviations and adaptive learning. E3 tests an information-congestion planner and a heterogeneous-provider objective frontier. Agreement across tasks is not transport: each task has its own nuisance-preservation and external support gate.

Our evaluation has five main findings.

1. Public menus identify a price-manipulation surface but not realized flow. The immutable panel contains 3,182,780 endpoint observations and 3,274 snapshots. For GLM-5.2, resetting three active quotes to the benchmark removes 18.62 percentage points of conditional shadow share. Its rank slope is linear-compatible; non-GLM active repricing is sparse. Neither proposed asymptotic regime is empirically established.
2. In a separate synthetic benchmark, an exact signal-order intervention validates a narrow Hansen–Misra–Pai-style mechanism [Hansen et al., 2021]. In a focal two-UCB prisoner's-dilemma game, preserving common signal order increases exploration coupling, joint high-price play, and buyer price. The two-seed factorial has 1,440 rows and 720 paired cells; the buyer-price effect is $+0.118$. The effect disappears under UCB/epsilon/static mixtures (-0.00047). The executable verdict is therefore negative on transport.
3. Hardening an adaptive score with lagged inputs, commitment epochs, operator-level caps, and one-sided share trust regions works mechanically. Across 4,878 historical menus it reduces mean maximum quote-attack gain from 0.226 to 0.030 and removes the measured identity-splitting gain. Yet the first UCB screen ends at profiles with normalized residual exploitability 3.929 versus 0.406 under inverse-square routing. Protecting a rule from one-step attacks can make its learning dynamics harder.
4. A conditional information-congestion benchmark has $k^* \asymp n^{(1+\alpha\beta)/(1+\alpha)}$ when signal effective rank scales as n^{β}. Its integer optimizer is reproduced, but the paired bandit allocation effect is null. Public rank diagnostics also reject the motivating GLM-minority/non-GLM-linear split.
5. A heterogeneous structural game makes mechanism objectives explicit. Across 108 policies, the welfare-selected rule attains welfare 208.44 and router revenue 4.88; the revenue-selected rule attains revenue 7.78 but welfare 83.99 and negative user utility. Zero unilateral grid regret does not eliminate profitable bounded pair deviations. These are declared scenarios, not live estimates or evidence of collusion.

The contribution is methodological and executable. Existing multi-agent frameworks provide general APIs [Terry et al., 2021, Lanctot et al., 2019]; algorithmic-pricing work demonstrates emergent coordination in stylized demand systems [Calvano et al., 2020]. We add request-level inference-market accounting, mechanism plugins, empirical property gates, and negative transport results.

2 The Strategic Routing Environment

2.1 Timing and primitives

There are providers $i \in \{1, \dots, n\}$. Provider i has private technology

$$\xi_i = (c_i, K_i, F_i, \rho_i, L_i),$$

where c_i is attempt cost, K_i physical capacity, F_i capital cost per slot, ρ_i reliability, and L_i base latency. At epoch t, it chooses

$$a_{it} = (p_{it}, \kappa_{it}, z_{it}),$$

where $p_{it} > 0$ is its quote, $\kappa_{it} \in [0, 1]$ its admitted-capacity fraction, and z_{it} is active, degraded, or withdrawn. Effective capacity is the integer floor of $K_i\kappa_{it}$ times a degradation multiplier.

A workload specifies input and output tokens, delivered value v, failure loss ℓ_F, latency cost ℓ_L, fallback latency, and a maximum number of attempts. A router observes public actions and whatever attributes its plugin is allowed to use, then returns an ordered list of eligible providers for each request. Each admitted attempt consumes capacity and cost even if it fails. Payment goes to the first successful provider. Capital cost is paid each epoch, including when capacity is withdrawn.

2.2 Exact settlement

For request r, let $j(r)$ be the serving provider, if any; $A(r)$ the attempted providers; P_r its payment; $C_r = \sum_{i \in A(r)} c_i$ resource cost; and T_r total latency. User utility and pre-capital welfare are

$$u_r = \begin{cases} v - P_r - \ell_L T_r, & j(r) \neq \varnothing, \\ -\ell_F - \ell_L T_r, & j(r) = \varnothing, \end{cases} \tag{1}$$

$$w_r = \begin{cases} v - C_r - \ell_L T_r, & j(r) \neq \varnothing, \\ -\ell_F - C_r - \ell_L T_r, & j(r) = \varnothing. \end{cases} \tag{2}$$

Provider profit is revenue minus attempt and capital costs.

Theorem 1 (Settlement reconciliation)**.** *For every seed, action profile, demand realization, capacity profile, router plugin, and fallback path generated by the kernel,*

$$W_t = U_t + \sum_i \pi_{it}. \tag{3}$$

No reliability, independence, or nonbinding-capacity assumption is required.

The identity is enforced in tests, as are capacity conservation, nonnegative payments, attempted-provider ordering, and deterministic replay. If a router fee or SLA transfer is added, it enters both sides explicitly.

2.3 Parallel RL interface and information boundary

The environment exposes

$$\texttt{reset(seed)} \to \{i : o_{i0}\}, \qquad \texttt{step(actions,demand)} \to (o', r, done, info, result).$$

The observation o_{it} contains epoch, demand, public quotes, and i's own profit, attempts, successes, technical failures, and capacity rejections. It does not contain rival technology ξ_{-i}. The complete market result remains available to an evaluator. This separation prevents a common simulation error: training agents on planner-only data and then interpreting their behavior as a market equilibrium.

The API is dependency-light and can be adapted to PettingZoo's parallel interface. It deliberately treats one epoch as a simultaneous provider move. Within-epoch routing and fallback occur after actions are committed; router state advances exactly once per epoch.

2.4 Mechanism and strategy plugins

The baseline inverse-price router uses $x_i \propto p_i^{-\eta}$. This is a counterfactual mechanism, not an empirical claim about OpenRouter. Adaptive routers may additionally use reliability or quality. The common interface writes

$$x_i(p, \alpha) = \frac{p_i^{-\eta} e^{\alpha_i}}{\sum_j p_j^{-\eta} e^{\alpha_j}}, \qquad p_i^{\text{eff}} = p_i e^{-\alpha_i/\eta}. \tag{4}$$

Thus every non-price score is a price-equivalent subsidy or penalty. The environment can reveal α_i to the provider, reveal only its realized attempts, or delay score feedback; the information policy is part of the router plugin and scenario hash. If $\alpha_i = h_i(\log p_i)$, negative own-price elasticity is $(1 - x_i)(\eta - h_i')$, making score attenuation or amplification of undercutting directly testable. The hardened rule includes five defenses: lagged smoothed inputs, provider- specific leave-one-out parameters, multi-epoch commitments, operator-neutral exploration and share caps, and a one-sided log-share trust region. Allocation gains are capped; providers are not insured against losing share after raising price or quality risk.

Strategies share the same action interface. Static policies reproduce fixed menus; species policies reproduce descriptive price regimes; Q and UCB policies learn from own reward; and scripted adversaries search quote, capacity, outage, coalition, and identity-splitting actions. A strategy name is never treated as a claim about a live provider.

3 Property Testing and Transport

Definition 1 (Property ladder)**.** *A simulator claim at level k may use evidence from levels $1, \ldots, k$ but may not inherit claims from a higher level:*

1. *accounting and transition invariants;*
2. *information-set and intervention fidelity;*
3. *held-out empirical moments;*
4. *adversarial and heterogeneous-strategy robustness;*
5. *prospective transport to owned or partner routing outcomes.*

Level 1 is exact software validation. Level 2 asks whether an intervention changes only its declared object. Level 3 asks whether the environment preserves properties visible in held-out data. Level 4 tests whether a result survives agents unlike the focal learner and profitable deviations outside its path. Level 5 is required for a live mechanism or welfare claim.

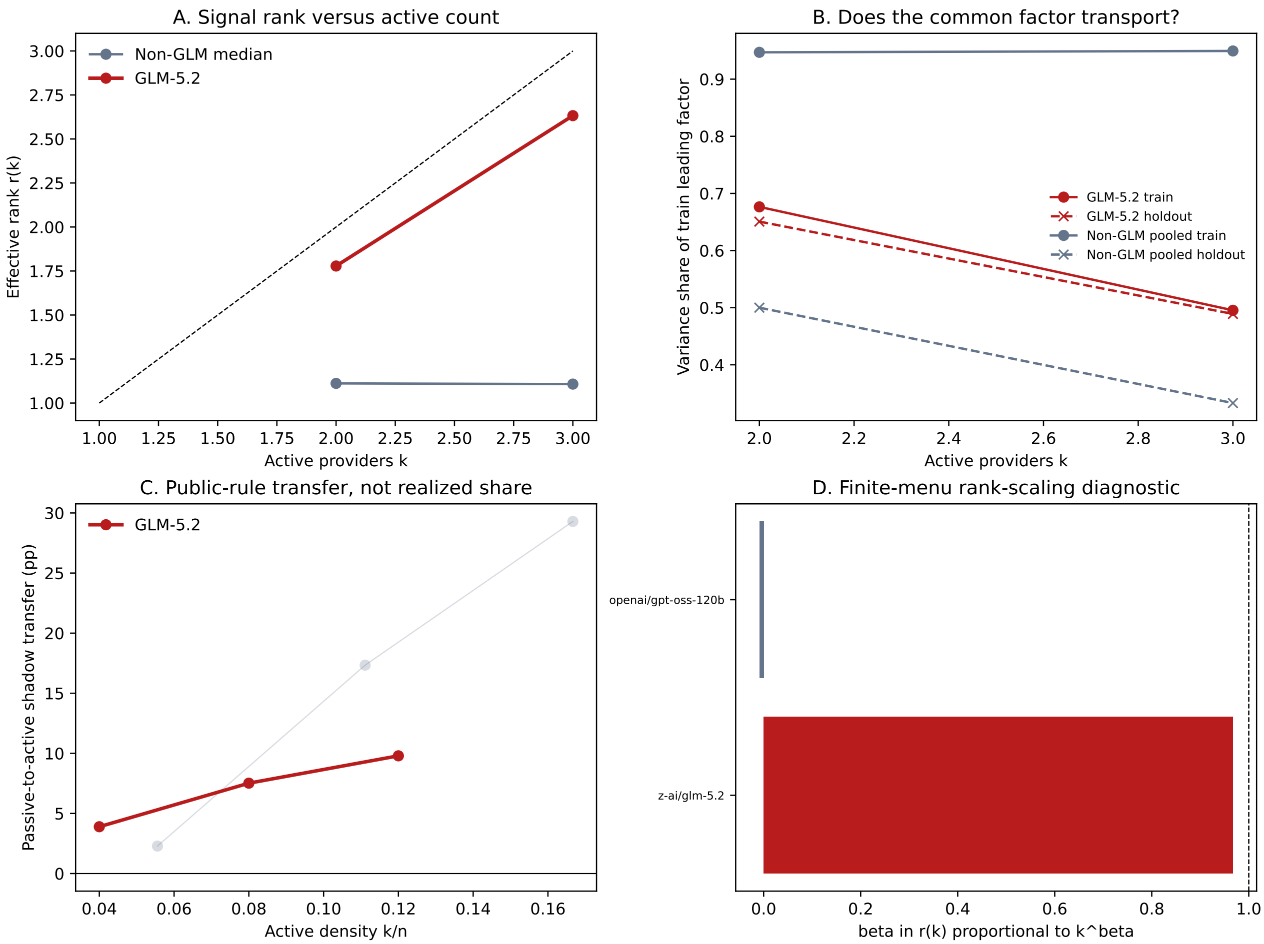

Figure 1: Level-3 public-market property test. GLM-5.2 has a small active set and large public-rule shadow transfer, but its rank slope is linear-compatible. Non-GLM activity is sparse and its only low-rank ladder fails temporal transport. The slopes are finite-menu diagnostics, not asymptotic estimates or realized shares.

3.1 Current market properties

The immutable hosted revision `f4023bdad64b6ad468741a5a3b1f3afb7af40dd0` spans 7–22 July 2026 and contains 3,182,780 endpoint observations, 3,274 five-minute snapshots, 321 model identifiers, and 72 provider labels. Free routes are excluded from price-share analysis, leaving 71 paid markets with stable menu support. A clean integrity run completes 29 registered public modules with zero failures and never queries prospective owned outcomes.

GLM-5.2 has three pre-holdout active undercutters—Novita, StreamLake, and Inceptron—in a median 25-provider menu. Their observed inverse-price shadow share is 27.61%; resetting their quotes to the author/menu benchmark gives 8.99%, a passive-to-active transfer of 18.62 percentage points. This is a deterministic rule counterfactual, not realized flow. Its nested effective-rank slope is 0.967 with block-bootstrap interval $[0.516, 1.219]$ at one hour and is linear-compatible at six and 24 hours as well. Across 70 non-GLM markets, 90% have zero active undercutters and the median active density is zero. The only non-GLM three-provider rank ladder loses its dominant training factor in the holdout. The attractive GLM-$o(n)$/non-GLM-$\Omega(n)$ contrast therefore fails level 3.

The level-5 companion is an assignment-first GLM-5.2 campaign. Thirty-six covered default choices over 18 blocks estimate a public-price exponent of 1.735 with profile interval $[0.31, 3.16]$. The focal active pair is selected 2/36 times versus 13.60% predicted shadow share. In 15 complete paired blocks, default routing selects the cheapest provider 0% of the time and explicit price sorting selects it 80% of the time; the block-bootstrap contrast is $[60, 100]$ percentage points. This is a direct action-interface effect for one account. The registered gate remains 800 choices, 100 blocks, seven days, and 90% candidate coverage. The latent-score fit has only 30 choices in 15 blocks and is not run.

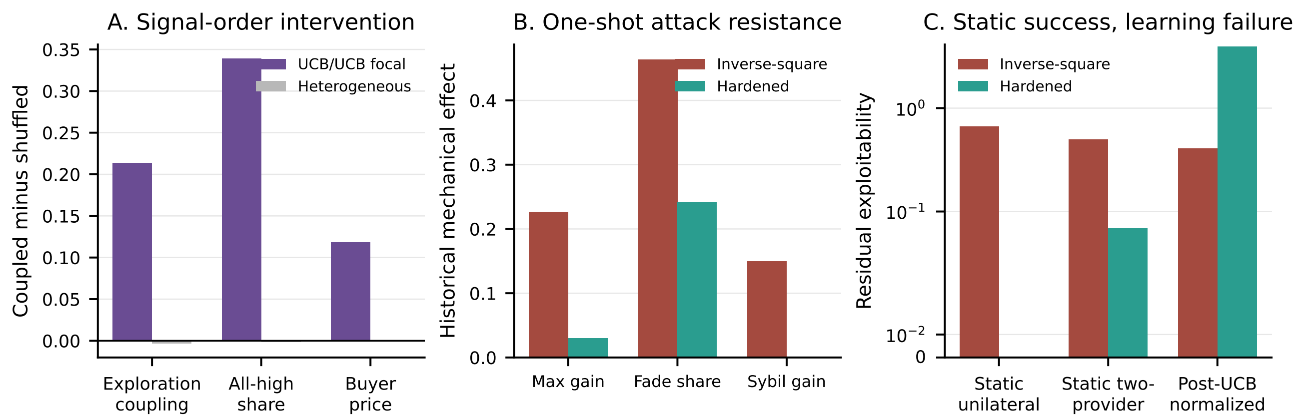

Figure 2: Property tests expose conditional success. Panel A: exact marginal-preserving signal intervention; focal UCB/UCB effects vanish across heterogeneous strategies. Panel B: the hardened adaptive router reduces mechanical one-step attacks on historical menus. Panel C: finite-grid static audits pass, but normalized post-UCB exploitability is higher. Panels B–C are simulator or menu-replay quantities, not live provider profit.

3.2 Intervention fidelity

The signal experiment starts from an $T \times n$ signal matrix Z. Its treatment arm preserves the original order; its control independently permutes each provider column.

Proposition 1 (Exact marginal preservation)**.** *For every provider i and every measurable function h,*

$$\sum_{t=1}^{T} h(Z_{ti}) = \sum_{t=1}^{T} h(\widetilde{Z}_{ti}). \tag{5}$$

Thus the intervention preserves each finite-sample marginal multiset exactly and changes only cross-provider and temporal ordering.

Paired seeds reuse action-policy RNG and structural primitives. The intervention therefore identifies a signal-order effect inside the simulator. It does not identify the signals, beliefs, or algorithm of a real provider.

4 Experiments

4.1 E0: price manipulation times latent scoring

For every provider below a model-author benchmark, E0 replaces only its quote with that benchmark. Let $\Delta_i(0)$ be the resulting price-only share gain and $\Delta_i(\alpha)$ the gain under equation (4). The interaction

$$\Gamma_i = \Delta_i(\alpha) - \Delta_i(0) \tag{6}$$

is negative when scoring attenuates undercutting and positive when it amplifies it. The public panel identifies $\Delta_i(0)$ mechanically. The prospective owned-choice panel estimates α; until its gate, Γ_i is missing, not zero. Simulator transport requires matching the sign and magnitude of Γ_i, not merely reproducing the public quote distribution.

This task also supplies adversarial policies: unilateral benchmark undercutting, fading a cheap quote after allocation, identity splitting, quality shading, and capacity withdrawal. A router passes only if it lowers these gains without violating held-out completion, fidelity, entry, and learning-regret gates.

4.2 E1: common signals and heterogeneous learners

E1 is an adversarial learning benchmark, not a proposed explanation of E0's cross-model shadow-share time series. Its memory and signal processes are declared simulator primitives. A live-market interpretation requires a separate transport experiment on lagged routing scores and quality outcomes.

Two providers choose low price 0.65 or high price 1.00 at marginal cost 0.20. With router exponent 5, the one-period payoffs satisfy the prisoner's-dilemma ordering

$$T = 0.4032 > R = 0.4000 > P = 0.2250 > S = 0.0832.$$

The factorial varies signal-to-noise, common correlation, exponentially weighted reward memory, router exponent, learner mixture, and seed. UCB/UCB is the focal mechanism; UCB/epsilon, epsilon/epsilon, and UCB/static are transport screens. Each coupled cell is paired with its marginal-preserving shuffle.

The successful hosted run contains 1,440 factorial rows and 720 paired cells. In the focal positive-correlation, SNR-at-least-two, exponent-five subset, coupled signals raise exploration-action correlation by 0.214, joint high-price share by 0.339, and buyer price by 0.118. Buyer price increases in 69.4% of focal cells. The primary focal sign screen passes.

The heterogeneous screen fails. Its corresponding effects are -0.0030, -0.0015, and -0.00047; buyer price increases in 49.1% of cells. Across all paired cells the mean buyer-price effect is 0.0131 and the positive share is 49.9%. We therefore report a causal mechanism example and a failed universality claim. A simulator that published only the focal UCB panel would give the wrong design impression.

4.3 Failed external transport audit for E1

This audit does not connect E1 to E0's market-share accounting. The current assignment-first HMP monitor contains 120 assignments, attempts, and covered choices with complete menu coverage and assignment integrity. It contains no qualifying public multiplicity event. Consequently only the exact price-path identity passes; routing-SNR, elasticity-wedge, buyer-harm, and familywise inference gates do not run. The score-memory monitor has 26 choices in 13 blocks over 0.55 days and also remains accruing. No signal-coupling, memory, communication, or collusion claim transports. A fixed property gate prevents the simulator's focal UCB result from becoming optional live-market language.

4.4 E2: static defense versus adaptive learning

The adaptive-router screen uses 4,878 eligible historical menus, 652,680 historical quote-attack rows, 200 strategic menus, 7,200 cost-by-capacity-by-rule cells, 32 sequential-response cells, 320 UCB rows, and 40 bounded Q-learning runs. All artifacts share one immutable input revision.

Against inverse-square routing, hardening reduces mean maximum allocation gain from 0.2262 to 0.0300, share captured by a 25% fading quote from 0.4638 to 0.2422, and two-identity combined-share gain from 0.1499 to zero. Model-day cluster intervals exclude zero in the favorable direction. Finite-grid mean unilateral exploitability falls from 0.6664 to zero; mean two-provider exploitability from 0.4976 to 0.0690. All eight hardened sequential best-response cells converge with zero residual gain within the declared grid.

The learning screen reverses the normalized ranking. Post-UCB residual exploitability is 3.9291 under hardening and 0.4057 under inverse-square, a 9.68 ratio. Absolute gains are much closer (5.89×10^{-4} versus 3.64×10^{-4}), so lower provider profit contributes to the normalized failure. The original UCB seed rows were deterministic duplicates; the correct independent unit is eight menus. A later stochastic engineering audit restores distinct seed paths and finds lower hardened absolute deviation gain on seven of eight menus, but still fails the frozen normalized gate. Neither result is confirmatory.

The design lesson is concrete. A one-sided trust region can prevent a provider from suddenly gaining traffic by undercutting, yet introduce committed state that learners estimate slowly. Static strategy-proofness on a grid does not imply low learning regret or fast convergence in a nonstationary repeated game.

4.5 E3: information congestion and objective frontiers

E3 asks two questions that E0–E2 leave open: how many providers should adapt to correlated information, and which router objective should select among their best responses? Let Σ_n be the covariance matrix of provider signals and define effective rank

$$r_n = \frac{(\operatorname{tr}\Sigma_n)^2}{\operatorname{tr}(\Sigma_n^2)}. \tag{7}$$

For k adaptive providers, declare the planner benchmark

$$V_n(k) = \frac{bk}{n} - c\left(\frac{k}{n}\right)^2\left(\frac{k}{r_n}\right)^{\alpha}, \qquad b, c, \alpha > 0. \tag{8}$$

The benefit is local allocation improvement; the loss is a scenario for common-mode overreaction. It is not identified from quote covariance.

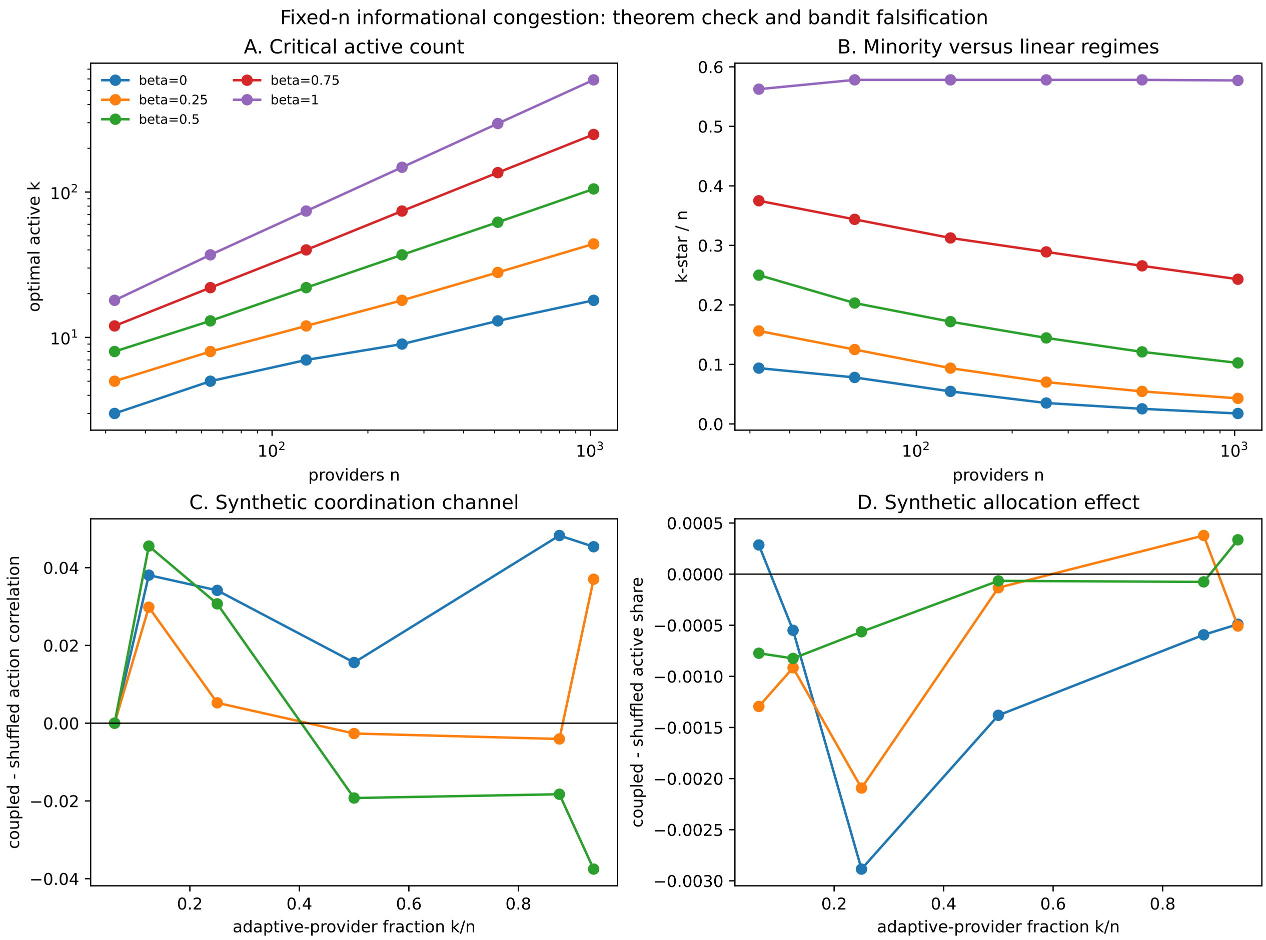

Figure 3: E3 information-congestion benchmark. Panels A–B recover the conditional optimizer. Panels C–D show the paired behavioral falsification: action correlation can move without a distinguishable active-share response. Numerical theorem recovery is not mechanism transport.

Theorem 2 (Conditional information-congestion threshold)**.** *Before clipping to* $[0, n]$*, the unique continuous maximizer of equation* (8) *is*

$$k_n^\circ = \left(\frac{bnr_n^\alpha}{c(2+\alpha)}\right)^{1/(1+\alpha)}. \tag{9}$$

The integer optimum is adjacent. If $r_n \asymp n^\beta$*, then*

$$k_n^* = \Theta\Big(n^{(1+\alpha\beta)/(1+\alpha)}\Big). \tag{10}$$

Thus the efficient adaptive fraction vanishes for $\beta < 1$ *and can remain constant for* $\beta = 1$*.*

SM3 holds n fixed within every comparison. Across $\beta = 0, .25, .5, .75, 1$, fitted integer-optimum exponents differ from equation (10) by at most 0.006. A paired bandit intervention then preserves every provider's marginal signal multiset while changing common order. It can change action correlation, but its active-share effects are -0.00116, -0.00086, and -0.00038 for $\beta = 0, .5, 1$, with all seed-clustered intervals containing zero. The environment therefore validates the optimizer and rejects the declared behavioral microfoundation.

SM4 declares ten technologies: an author, two anchor resellers, two reserved-capacity providers, three spot-dependent startups, and two custom-quality providers. Providers play bounded cyclic price best responses against 108 router rules spanning price exponent, quality and reliability weights, exploration, correlated-group caps, and take rates. Every outcome reconciles welfare, user utility, provider profit, and router fees; 104 rules converge.

The inverse-square price-only baseline has welfare 137.29, router revenue 5.99, quality-adjusted successes 111.52, ten viable providers, and served-share HHI 0.125. The welfare-, quality-, and user-selected rule has welfare 208.44, quality-adjusted successes 149.96, revenue 4.88, eight viable providers, and HHI 0.281. The revenue-selected rule has revenue 7.78

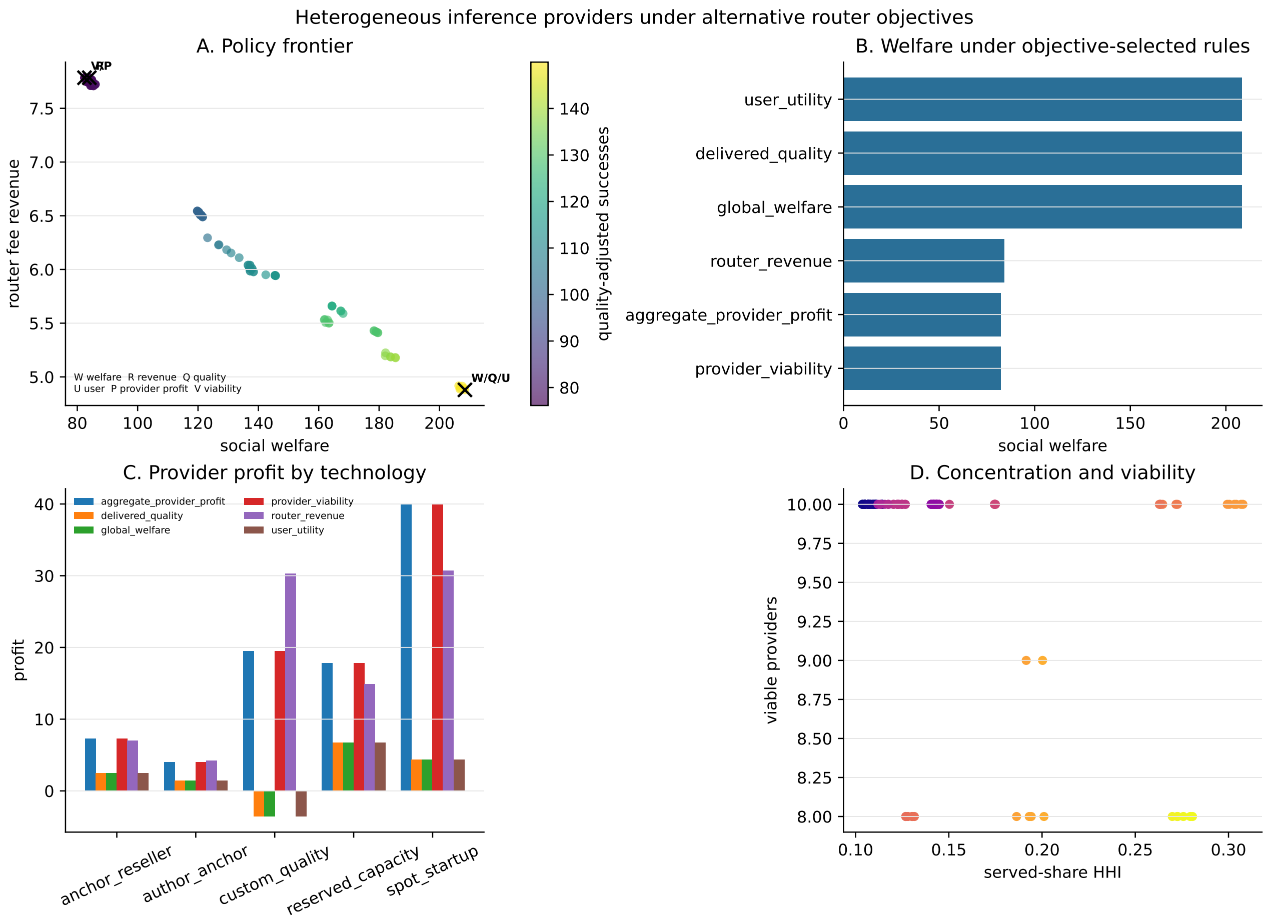

Figure 4: E3 heterogeneous-provider objective frontier. Rules selected for welfare, router revenue, delivered quality, user utility, provider profit, and viability occupy different regions. Technologies, demand, and value are declared scenarios; the figure is not a live-market welfare estimate.

but welfare 83.99 and negative user utility. Selected rules have zero unilateral grid regret. Nevertheless, the largest bounded same-type pair deviation gains 1.87 under the welfare rule and 8.88 under the revenue rule. These are susceptibility tests, not evidence of coordinated conduct.

5 Mechanism Evaluation Without Welfare Theater

The kernel computes user utility, provider profit, and social surplus for every declared scenario. Those numbers are internally exact and externally conditional on workload value, provider cost, and capacity. Current public data identify none of these primitives well enough for a market-wide welfare estimate.

We therefore recommend a robust design problem. Let θ parameterize score weights, commitments, exploration, caps, and fees; Ω be the set of cost, quality, capacity, and learning models consistent with observed properties. The evaluator solves

$$\min_{\theta} \sup_{\omega \in \Omega} R_U(\theta, \omega) + \lambda_1 E_1(\theta, \omega) + \lambda_2 E_2(\theta, \omega) + \lambda_V V(\theta, \omega), \tag{11}$$

subject to completion, concentration, budget, and delivered-capacity floors. R_U is user regret; E_1 and E_2 are unilateral and two-provider exploitability; V is constraint violation. Results should be reported as a Pareto frontier, with cost/value sensitivity, rather than a single welfare point.

We instantiate four score families rather than treating "better routing" as a scalar label. The user-cost score minimizes payment plus latency and failure; the welfare score replaces payment by bounded resource cost and adds delivered value; the router-revenue score maximizes fees net of retries and SLA credits; and the quality score maximizes blinded task fidelity subject to spend and completion constraints. Each arm reports user utility, provider profit, router revenue, welfare bounds, quality, concentration, and exploitability. An arm can dominate on one vector and fail on another.

The outer game also varies provider technology. Owned or reserved-capacity providers pay fixed capital cost and have low short-run marginal cost; spot-dependent providers have flexible entry and scarcity-sensitive cost; anchor adopters may keep displayed price fixed; and technology-rent providers may invest in kernels or hardware. Entry, capacity, and quality investment are chosen before the repeated pricing game. A mechanism improvement must survive best responses and two-provider coalitions for each technology mixture, not just reallocate a frozen menu.

The simulator suggests a mechanism stack rather than a scalar score. First, use execution-contingent capacity commitments instead of advertised capacity or provider identity. Second, bound aggregate attempt probability assigned to providers with correlated quote innovations while preserving an operator-neutral exploration floor. Third, estimate quality and reliability out of sample and pay explicit SLA credits. Fourth, separate router fees from selected spend so the revenue objective is not silently inserted into the user score. Finally, publish a small Pareto menu of user-cost, welfare, revenue, and quality policies rather than pretending their weights are common knowledge.

Technology stress tests show why robustness must precede deployment. The welfare-selected rule falls from 208.44 to worst-case welfare 147.65 under reserved-capacity shortfall, reliability shocks, spot-cost spikes, and quality compression. The revenue and provider-viability selections have worst-case welfare below 80. The next live experiment should therefore randomize default delegation, explicit price sorting, and one predeclared score within model–prompt–time blocks; measure spend, completion, latency, and blinded task fidelity; and report objective-specific outcomes over a preregistered cost/value box. Capacity commitments require a separate provider-facing trial.

6 Related Work

General multi-agent interfaces and game-learning suites prioritize broad domain coverage [Terry et al., 2021, Lanctot et al., 2019]. Our contribution is a market-specific settlement layer and an empirical transport standard. Algorithmic-pricing experiments show that independent learners can sustain high prices [Calvano et al., 2020]. Hansen, Misra, and Pai identify a distinct channel through correlated price experiments [Hansen et al., 2021]; our exact signal-order intervention isolates that channel and shows its learner dependence.

Brown and MacKay study heterogeneous pricing algorithms and repricing frequency [Brown and MacKay, 2023]. Our persistent quote regimes are compatible with that motivation but do not identify algorithms. Johnson, Rhodes, and Wildenbeest show how platform prominence changes algorithmic price competition [Johnson et al., 2023]; our adaptive-rule experiments add hidden capacity, fallback, and learning-speed failures. Model-routing benchmarks choose among models [Chen et al., 2023, Ong et al., 2024, Hu et al., 2024]; we hold the model label fixed and study competition among execution providers.

7 Limitations, Ethics, and Conclusion

The public panel is measured in weeks. Owned probes cover one account and workload distribution. Public enforcement is aggregated. Provider cost, market-wide demand, user value, private rebates, and deliverable capacity are unobserved. UCB, epsilon-greedy, Q-learning, and static policies span only a small subset of strategic behavior. Grid exploitability is not unrestricted dynamic exploitability. The information-congestion loss is a declared scenario, not identified by effective rank; its bandit allocation transport is null. SM4 technologies are stress cases rather than fitted provider types. The hardened-rule and owned-routing screens are mixed or accruing.

Paid probes are assignment-first, minimal-token, budget-capped, and remotely rate-limited. Public artifacts remove request-level rows. The research does not send adversarial load, infer prompt contents, or identify provider employees. Provider labels are descriptive regimes, not allegations. We do not claim collusion, communication, dumping, or literal front-running.

The central finding is that routing-market simulation needs a refusal mechanism. Public menus reveal a large price-induced shadow-share surface but falsify the proposed GLM/non-GLM scaling story. The conditional congestion optimizer is easy to reproduce numerically; its behavioral channel fails. A hardened router resists one-step manipulation and fails its first learning gate. Objective- selected equilibria disagree sharply even before model uncertainty, and unilateral stability does not remove pairwise susceptibility. Owned choices are required for hidden scoring and are still gated. The property ladder makes each failure executable. That separation—not a simulated welfare number or a conduct label—is what turns a strategic routing environment from a storytelling device into a scientific instrument.

A Proofs

Proof of theorem 1. For a successful request, user utility plus the serving and attempted providers' variable profit is $(v - P_r - \ell_L T_r) + (P_r - C_r) = v - C_r - \ell_L T_r = w_r$. For a failed request it is $(-\ell_F - \ell_L T_r) - C_r = w_r$. Sum over requests and subtract each provider's epoch capital cost on both sides. □

Proof of proposition 1. For each i, $\widetilde{Z}_{\cdot i}$ is a permutation of $Z_{\cdot i}$. Summation is invariant to permutation. □

Proof of theorem 2. Equation (8) equals $bk/n - ck^{2+\alpha}/(n^2 r_n^\alpha)$. Its derivative vanishes only at the displayed k_n°, and its second derivative is strictly negative for $k > 0$. Strict concavity places the integer optimum at an adjacent integer. Substituting $r_n \asymp n^\beta$ gives equation (10). □

B Reproducibility and Evidence Ledger

The RL wrapper is `src/orcap/market_env/rl_env.py`; its tests verify private-information boundaries, reward reconciliation, horizons, invalid calls, and exact reset replay. A documented four-epoch example at `examples/strategic_routing_demo.py` is itself tested for byte-equivalent seeded output and settlement reconciliation; the adjacent environment guide specifies the observation contract and custom-router path. Kernel tests additionally cover capacity, failure, fallback, transfers, and stable random substreams. Signal tests verify exact column multisets and paired interventions. Adversarial tests cover quote monotonicity, share caps, commitment state, identity grouping, unilateral grids, and coalition enumeration.

The current public input is immutable Hugging Face revision `f4023bdad64b6ad468741a5a3b1f3afb7af40dd0`. A clean integrity run executes 29 registered public modules with zero failures and explicitly excludes prospective owned outcomes. WF20 produces the paid-market census, nested-rank curves, block bootstrap, and benchmark-reset shadow shares. The current owned figures come only from aggregate monitor artifacts; request-level outcomes are not inspected or published before their gates.

SM3 is generated by `src/orcap/analysis/sm3_informational_congestion.py`; tests cover effective rank, analytic and integer optima, fixed-n scaling, marginal-preserving paired signals, and seed-clustered intervals. SM4 is generated by `src/orcap/analysis/sm4_multiobjective_router_game.py`; tests cover fixed-provider menus, capacity redistribution, welfare accounting, bounded best responses, objective selection, technology stress, pair deviations, and JSON serialization. The older E1 and E2 bundles retain their recorded code, input, seed, and artifact hashes in `papers/neurips/hosted-evidence.json`. No public shadow share, latent score, provider technology, cost, or value is fitted into SM3 or SM4. The full boundary is recorded in the dated information-congestion evidence audit adjacent to the venue folders.

C Environment Card

Agents and actions. The agents are inference providers. A simultaneous action is a positive quote, an admitted-capacity fraction in $[0, 1]$, and an availability state. The core environment does not let a provider change its private marginal cost, physical capacity, reliability, or base latency during an episode. Experiments that model investment do so in an outer stage and create a new immutable provider specification.

Observations and rewards. Every provider sees the epoch, realized demand, the public quote vector, and its own preceding profit, attempts, completions, failures, and capacity rejections. The default reward is own profit after variable and installed-capacity cost. Alternative rewards must be named by an experiment; the environment never hands social welfare to a provider by default. The evaluator sees request outcomes, user utility, provider profit, resource cost, payment, and conditional welfare.

Transition and termination. Within an epoch, the router orders candidates, capacity clips attempts, reliability draws determine technical failure, and fallback continues to the declared limit. The episode ends at a fixed horizon; there is no outcome-based early stopping. Calling `step` after termination is an error. Resetting with the same seed exactly replays the same transition under identical actions.

Randomness and pairing. A stable hash of base seed, epoch, request, event type, and provider creates disjoint random substreams. Mechanism comparisons reuse the same base seed so a routing change does not silently redraw reliability. Signal interventions have a separate, recorded seed and assert column-multiset equality before running.

Known failure modes. Expected-allocation training removes routing noise and can create duplicate seed paths. Normalized exploitability is unstable near zero provider profit. A finite quote grid can miss continuous or history-dependent deviations. Public menus cannot calibrate hidden eligibility or market-wide demand. Static action robustness does not imply learning convergence. The released screen reports all five failures rather than treating them as tuning opportunities.

Minimum reporting standard. A routing-market experiment should publish: code and scenario hashes; action and observation schemas; provider information boundaries; seed and pairing policy; router state-transition timing; transfer reconciliation; capacity and fallback semantics; calibration/holdout split; algorithm families; global and coalition deviation class; absolute and normalized exploitability; failed transport gates; and a statement of which welfare primitives are assumed rather than measured.

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